AI ASSISTED CODING

ASSIGNMENT-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

Instructions:

• Prompt AI to convert a Python class with attributes and methods into equivalent Java code.

• Verify that constructors, method signatures, and data types are properly handled.

• Run both versions (Python & Java) to compare expected outputs.

Expected Output:

• Equivalent working Java class translated from Python.

```python
# create a python code using class with attributes and methods
class BookCollection:
    def __init__(self):
        self.books = []

    def add_book(self, book: str) -> None:
        self.books.append(book)

    def remove_book(self, book: str) -> None:
        if book in self.books:
            self.books.remove(book)
        else:
            print(f"Book '{book}' not found in the collection.")

    def list_books(self) -> list:
        return self.books
# Example usage:
if __name__ == "__main__":
    bc = BookCollection()
    bc.add_book("To Kill a Mockingbird")
    bc.add_book("1984")
    print(bc.list_books())  # Output: ['To Kill a Mockingbird', '1984']
    bc.remove_book("1984")
    print(bc.list_books())  # Output: ['To Kill a Mockingbird']
    bc.remove_book("The Great Gatsby")  # Output: Book 'The Great Gatsby' not found in the collection.
```

```java
# convert the above code to Java language
import java.util.ArrayList;
import java.util.List;
public class BookCollection {
    private List<String> books;

    public BookCollection() {
        this.books = new ArrayList<>();
    }

    public void addBook(String book) {
        books.add(book);
    }

    public void removeBook(String book) {
        if (books.contains(book)) {
            books.remove(book);
        } else {
            System.out.println("Book '" + book + "' not found in the collection.");
        }
    }
    public List<String> listBooks() {
        return books;
    }

    public static void main(String[] args) {
        BookCollection bc = new BookCollection();
        bc.addBook("To Kill a Mockingbird");
        bc.addBook("1984");
        System.out.println(bc.listBooks()); // Output: [To Kill a Mockingbird, 1984]

        bc.removeBook("1984");
        System.out.println(bc.listBooks()); // Output: [To Kill a Mockingbird]

        bc.removeBook("The Great Gatsby"); // Output: Book 'The Great Gatsby' not found in the collection.
    }
}
```

```
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe c:/Users/gudah/OneDrive/Documents/AIAC/lab_assignment-19-1.py
['To Kill a Mockingbird', '1984']
['To Kill a Mockingbird']
Book 'The Great Gatsby' not found in the collection.
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code>
```

```
Output                                                    Clear
[To Kill a Mockingbird, 1984]
[To Kill a Mockingbird]
Book 'The Great Gatsby' not found in the collection.

=== Code Execution Successful ===
```

Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

• Ask AI to translate a recursive C++ factorial function into Python.

• Validate correctness by testing with sample inputs.

• Ensure Python version follows Pythonic conventions.

Expected Output:

• A correct Python implementation equivalent to the original C++ factorial function.

```cpp
# write a c++ code to calculate factorial
#include <iostream>
using namespace std;
int factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    }
    return n * factorial(n - 1);
}
int main() {
    int number;
    cout << "Enter a number to calculate its factorial: ";
    cin >> number;
    cout << "Factorial of " << number << " is " << factorial(number) << endl;
    return 0;
}

# convert the above code to python language
def factorial(n):
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)
if __name__ == "__main__":
    number = int(input("Enter a number to calculate its factorial: "))
    print(f"Factorial of {number} is {factorial(number)}")
```

```
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe c:/Users/gudah/OneDrive/Documents/AIAC/lab_assignment-19-1.py
Enter a number to calculate its factorial: 5
Factorial of 5 is 120
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code>
```

Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in

Python.

Instructions:

• Provide AI with a SQL query (e.g., SELECT name, salary

FROM employees WHERE salary > 50000).

• Ask AI to generate equivalent Pandas code.

• Test the generated code on a sample DataFrame.

Expected Output:

• Equivalent Pandas query that retrieves the correct subset of data.

```
"""CREATE TABLE employees (
    employee_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    salary DECIMAL(10, 2) NOT NULL
);
INSERT INTO employees (name, salary) VALUES ('Alice', 60000);
INSERT INTO employees (name, salary) VALUES ('Bob', 45000);
INSERT INTO employees (name, salary) VALUES ('Charlie', 55000);
SELECT name, salary FROM employees WHERE salary > 50000;"""

import pandas as pd
data = {
    'employee_id': [1, 2, 3],
    'name': ['Alice', 'Bob', 'Charlie'],
    'salary': [60000, 45000, 55000]
}
df = pd.DataFrame(data)
result = df[df['salary'] > 50000][['name', 'salary']]
print(result)
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe c:/Users/gudah/OneDrive/Documents/AIAC/lab_assignment-19-1.py
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe c:/Users/gudah/OneDrive/Documents/AIAC/lab_assignment-19-1.py
     name  salary
0   Alice   60000
2 Charlie   55000
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code>
```

Task 4 – Real-Time Application: Multi-Language Snippet Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

• Converting a Python web scraping snippet into JavaScript for browser automation.

• Translating a Java method into C# for enterprise applications.

Instructions:

• Prompt AI to translate code between two chosen languages.

• Test both implementations to ensure they produce the same results.

• Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

• Working equivalent code snippets in at least two different programming languages.

```python
#generate a python code that print N natural numbers
from py_compile import main
N = int(input("Enter a number: "))
for i in range(1, N+1):
    print(i)

#Conver the above code to  C language
#include <stdio.h>

int main() {
    int N;
    printf("Enter a number: ");
    scanf("%d", &N);
    for (int i = 1; i <= N; i++) {
        printf("%d\n", i);
    }
    return 0;
}

#convert the above code to Go
package main
import "fmt"
func main() {
    var N int
    fmt.Print("Enter a number: ")
    fmt.Scan(&N)
    for i := 1; i <= N; i++ {
        fmt.Println(i)
    }
}
```

```
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code> & c:\python314\python.exe c:/Users/gudah/OneDrive/Documents/AIAC/code_translation.py
Enter a number: 5
1
2
3
4
5
PS C:\Users\gudah\AppData\Local\Programs\Microsoft VS Code>
```

Challenges: Differences in libraries, syntax (indentation vs braces), and execution models (sequential vs asynchronous).

Resolution: Used async/await in JavaScript, adjusted output formats, and verified functional equivalence.

Learning: Translation requires adapting to language paradigms and selecting equivalent libraries.